Sensorless Power Measurement for Personal Computers Using CMOS MEMS Magnetometers and Machine Learning

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***Abstract*— We present a novel, sensorless power meter for accurate and real-time power estimation of personal computers (PCs). Our innovative system integrates CMOS MEMS magnetic field sensors with machine learning-based IoT analysis, allowing power consumption to be measured without traditional sensors. At the core of our device is a CMOS MEMS search coil magnetometer, fabricated using the TSMC 1P6M 180nm process and calibrated against a commercial power meter. Coupled with Python software that monitors GPU utilization, CPU usage, and two temperature metrics, we developed a multivariate nonlinear regression (MNR) model with 59 terms to predict power consumption. After an initial training phase, our system can measure the power consumption of any PC sensorlessly, achieving less than a 2.95% discrepancy in average power measurements over 20 minutes. This scalable approach facilitates deployment across multiple PCs or servers in data centers, enabling real-time power reporting and carbon footprint calculations. By efficiently monitoring power usage in diverse computing environments, our method significantly enhances energy management and sustainability efforts.**

Keywords*— *Sensorless Power Measurement, CMOS MEMS Magnetometer, Machine Learning, Energy Efficiency.

I. Introduction

Accurate measurement of PC power consumption is critical for optimizing energy efficiency, reducing electricity costs, and minimizing carbon footprints. Conventional methods for power measurement typically involve hardware sensors, such as shunt resistors, Hall effect sensors, or power meters. These methods incur additional costs, require complex integration, and may not scale well for multiple devices. In contrast, sensorless power estimation offers a cost-effective and non-invasive alternative by eliminating the need for additional hardware and leveraging readily available system parameters. This makes sensorless methods particularly advantageous for environments with numerous PCs, such as data centers, where scalability, cost efficiency, and ease of deployment are priorities[1][2].The power consumption of a personal computer (PC) is closely tied to the performance and operating temperature of its Central Processing Unit (CPU) and Graphics Processing Unit (GPU). Both components dynamically adjust their voltage and frequency based on workload through techniques like Dynamic Voltage and Frequency Scaling (DVFS), which can lead to increased power usage during high-demand tasks. Elevated temperatures can cause thermal throttling, reducing performance and power consumption. Additionally, cooling systems must work harder to maintain optimal temperatures, leading to increased energy usage. Furthermore, the materials and designs of CPUs and GPUs contribute to power consumption through increased leakage current at higher temperatures. Overall, the interplay between workload, temperature, and power management strategies is critical for optimizing energy efficiency in PCs [3][4].

Several studies have explored power measurements of microprocessors using machine-learning approaches that rely on datasheet information [5]. In reference [6], a machine learning-based hierarchical power modeling approach for CPUs was presented, enabling accurate cycle-by-cycle power estimation at the microarchitecture level using low-complexity regressors and high-level activity features, validated on RISC-V cores with less than 3.6% error compared to gate-level power estimates. In the case of PCs, power assessment typically involves hardware solutions [7]. To the authors' knowledge, no prior research has introduced a semi-empirical model for estimating PC power consumption. Our approach uses CPU and GPU parameters, calibrated with a highly accurate power measurement system, to estimate power consumption. For the first time, a sensorless method is proposed that enables real-time power estimation using CPU and GPU state parameters, such as utilization and temperature, combined with a multivariate nonlinear regression (MNR) model. Additionally, we designed a highly sensitive CMOS MEMS search coil magnetometer using the TSMC 1P6M 180nm CMOS process to create an accurate measurement device and extract coefficients for power estimation.

II. WORKING PRINCIPLES

The proposed smart electrical power measurement system consists of two phases, as illustrated in the diagram shown in Figure 1. The power measurement system is implemented in two phases, as follows:

In the first phase, the electrical power consumed by the PC is measured using an in-house designed power meter. The data recorded from the power meter is synchronized with the performance parameters of the PC, as shown in Figure 1(a). There are more than 150 parameters that can affect the power consumption of the PC. After generating the correlation matrix with power consumption, the four most highly correlated parameters were selected for inclusion in the polynomial expansion to maintain model efficiency. Only four of these parameters were found to have a dominant effect on power consumption, as listed in Table 1. Introducing additional terms beyond these four would result in diminishing returns, as the improvement in accuracy is minimal. Through rigorous testing, the optimal degree for the polynomial terms was determined to be 6. While other power terms were evaluated, using 6 yielded the most accurate results without significantly increasing computational time. By utilizing the generalized linear regression method [8], we established the relationship between these four parameters and the estimated power (Pes), expressed in Equation 1. This equation consists of 59 terms, and its coefficients are detailed in Table 2.

Figure 1. (a) The training and coefficient extraction phase, (b) The evaluation phase of the proposed power measurement system.

In the second phase, the coefficients of Eqn. (1) are held constant while we measure the PC's power consumption using both our in-house developed power meter and the trained model. This phase is essential for evaluating the trained model's effectiveness and accuracy in estimating electrical power usage. By keeping the coefficients constant, we ensure that measurement variations result from differences between the two methods rather than fluctuations in the model parameters. The goal is to rigorously compare the model's outputs with the reliable readings from the power meter. This analysis helps quantify the model's accuracy and identify discrepancies. By systematically reviewing the results, we ensure precise power consumption estimates under various conditions, ultimately validating its robustness and applicability for future energy management applications.

$$
\begin{aligned}
P_{es} = &\sum_{j=1}^{4}\sum_{i=1}^{6} A_{ji}X_j^i + \sum_{i=1}^{4} B_i Ln(X_i) \\
&+ \sum_{k=1}^{3}\sum_{j=k+1}^{4}\sum_{i=1}^{5} C_{kji}X_k^i X_j^{6-i} \\
&+ D_{59}X_1X_2X_3X_4
\end{aligned} \tag{1}
$$

TABLE 1. LIST OF EXTRACTED PARAMETERS FOR PC POWER ESTIMATION.

X_1	GPU utilization [%]
X_2	CPU usage average [%]
X_3	CPU cores' temperature average
X_4	CPU GT cores' average temperature

TABLE 2. VALUE OF EXTRACTED COEFFICIENTS IN EQUATION (1).

A_{11}	-1.6786805	A_{43}	82.92042598	C_{143}	0.025737161
A_{12}	0.170616624	A_{44}	-1017.2399	C_{144}	-0.00056382
A_{13}	-0.00798242	A_{45}	-1499.5087	C_{145}	5.20291e-06
A_{14}	0.00015384	A_{46}	-1447.25599	C_{231}	-3.95865734
A_{15}	-1.1027e-06	B_1	-861.93459	C_{232}	0.728495642
A_{16}	1.27081e-09	B_2	253.9325027	C_{233}	-0.03013574
A_{21}	8.756308974	B_3	1896.682673	C_{234}	0.000537724
A_{22}	2.96066787	B_4	-575.494787	C_{235}	-4.3728e-06
A_{23}	0.026900507	C_{121}	1.39364e-10	C_{241}	8.662455444
A_{24}	-0.00175446	C_{122}	-5.0552e-10	C_{242}	-1.04986702
A_{25}	1.66632e-05	C_{123}	8.93501e-10	C_{243}	0.038496554
A_{26}	4.53197e-08	C_{124}	-5.1075e-10	C_{244}	-0.00063133
A_{31}	-7.9934e-10	C_{125}	1.42884e-10	C_{245}	4.74737e-06
A_{32}	-31.8009186	C_{131}	-4.38974261	C_{341}	843.8750386
A_{33}	168.1911253	C_{132}	0.499260051	C_{342}	-10.8233515
A_{34}	333.762958	C_{133}	-0.02355668	C_{343}	-591.174569
A_{35}	459.1104516	C_{134}	0.000495971	C_{344}	-815.309415
A_{36}	503.1029413	C_{135}	-4.6657e-06	C_{345}	-595.359989
A_{41}	420.6958048	C_{141}	3.131340902	D_{59}	-0.00233898
A_{42}	162.9304906	C_{142}	-0.4803459		

The in-house developed power meter consists of three main components, as shown in Figure 2(a). The micro search coil magnetometer (μSCM) was designed and fabricated using the TSMC 1P6M 180 nm RF general-purpose process. Detailed descriptions of the sensor design and fabrication of the μSCM can be found in our previous works [9]–[13]. When an electrical current passes through the power wire connected to the PC, a magnetic field is generated around the wire, as illustrated in Figure 2(b). This magnetic field is proportional to the current flowing through the wire, according to the Bio-Savart law [14]. The μSCM converts this magnetic field into an AC voltage, which is then amplified, rectified, and filtered by a low-pass filter within the electrical circuit designed on the PCB. The output voltage from the PCB is sent to an Arduino Yun module for digitization and data transfer to the PC. Notably, the in-house power meter was calibrated against a commercial power meter (NAPUI PM9805 Multifunction RS232 RS485 Communication Digital Power, Dongguan NAPUI Electronic Technology Co., Ltd., China) before conducting the experiments.

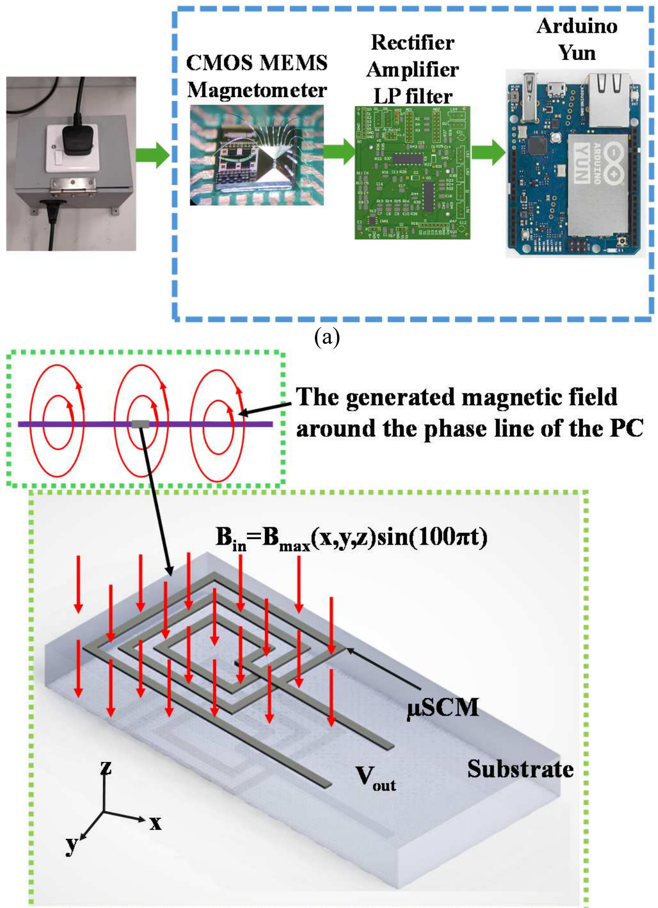

Figure 2. (a) The main building block of the in-house developed power meter, (b) the working principle of the micro search coil magnetometer, and the passing current measurement principle through the phase line connected to the PC.

Experiments were conducted on a Dell OptiPlex 7050 PC to evaluate the system's performance under various scenarios, including idle states (minimal CPU and GPU usage), moderate loads (e.g., web browsing and video playback), and high loads (e.g., 3D rendering and gaming). The experiments consisted of a 1,200-second training phase, during which power consumption was measured in four intervals to determine the 59 coefficients of Eqn. (1) using multi-purpose linear regression. The fitted results showed a maximum discrepancy of 7.92 watts compared to actual measurements, with average power values of 60.86 watts and 60.89 watts for the fitted and measured results, respectively, validating the model's accuracy. During the validation phase, GPU utilization peaked at 85–90%, and CPU usage varied between 10–80%, demonstrating the robustness of the proposed method in accurately predicting power consumption across diverse real-world PC tasks.

In the second phase of the experiments, the power consumption of the PC was measured using the power meter and calculated with Eqn. (1). Fig. 3(b) shows the comparison between measured and predicted power consumption over 1,200 seconds, divided into four steps. The maximum discrepancy between the measured and predicted power consumption was 8.88 watts, with average values of 70.94 watts for the measured power and 73.15 watts for the predicted power. The validation phase confirmed the accuracy of the predictions, indicating an error of less than 2.95 percent for the average power consumption. The raw data and the Python code for recording the data, as shown in Fig. 3, are available on github.com.

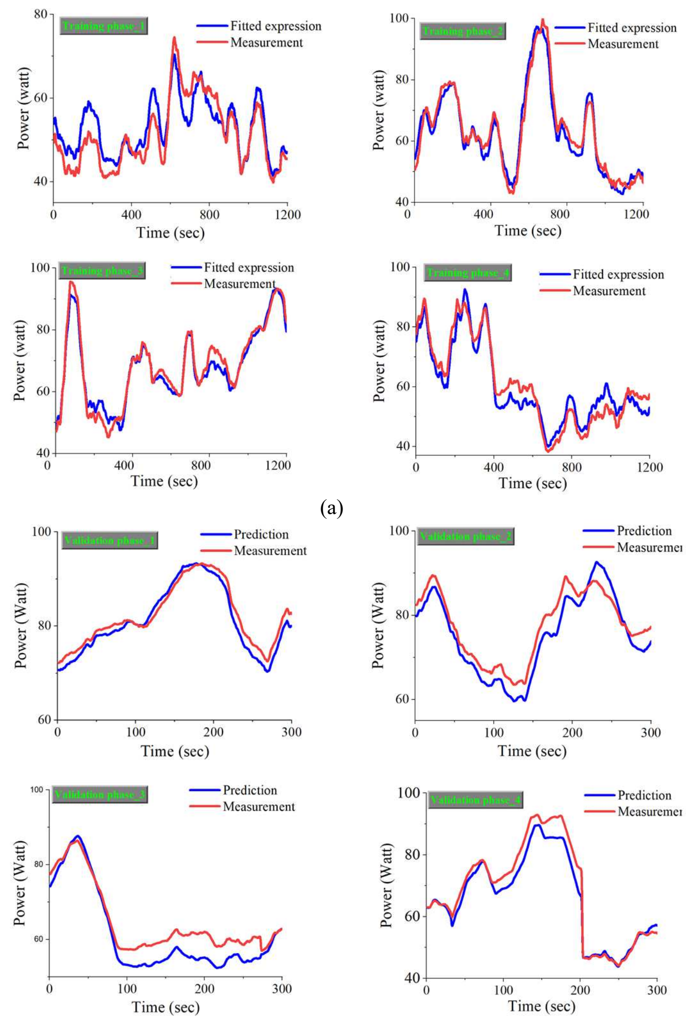

Figure 3. (a) Measurement results for the training phase, (b) Measurement results for the validation phase of the proposed power measurement system.

Table 3 compares the proposed method with existing solutions. Unlike traditional approaches that rely on hardware sensors or software tools, our hybrid method combines a CMOS MEMS magnetometer with a machine learning-based regression model. This enables sensorless power estimation with greater accuracy and scalability, ideal for data centers and multi-PC environments.

TABLE 3. COMPARISON BETWEEN THIS WORK AND PREVIOUSLY REPORTED WORKS.

Ref	Methodology	Measured parameters	Unique features
This work	CMOS MEMS magnetometer + software	Current, CPU, and GPU parameters,	Hybrid model for power prediction with accuracy higher than 2.95 percent for average power
[1]	Commercial power meter	Current, and Voltage	Analyzing the power consumption of PCs
[9]	CMOS MEMS magnetometer	Current	Theoretical analysis for the employed sensor design
[15]	Software	CPU, GPU temperature, fan speeds, and power consumption estimates	Hardware monitoring with logging capabilities
[16]	Software	CPU temperature, motherboard temperature, fan speeds	Hardware monitoring with logging capabilities
[17]	Software	GPU temperature, core clock speed, memory usage	Provides real-time monitoring with overclocking features

III. CONCLUSION

In this study, we developed a novel sensorless power measurement system for personal computers by leveraging a CMOS MEMS magnetometer and a multivariate nonlinear regression model. Our innovative approach utilizes four key performance metrics—GPU utilization, CPU average usage, CPU core temperature, and CPU GT core temperature—combined with software-driven analysis to estimate power consumption accurately. We established a robust relationship between these parameters and power usage by implementing a generalized linear regression model represented by a polynomial logarithmic expression. Experimental validation showed that the system achieves a maximum discrepancy of less than 2.95% in average power consumption across various PC operating scenarios, confirming its reliability and effectiveness. This method eliminates the need for traditional sensors, significantly reducing costs and enhancing scalability, especially in environments with multiple devices. Furthermore, the system enables real-time power estimates, marking a significant advancement in energy monitoring technology and holding considerable potential for applications in data centers, where efficient energy management is crucial for reducing operational costs and carbon footprints.

Future work will focus on integrating additional performance metrics, multi-modal deep learning models (such as OpenAI GPT-4o), and the open-source Sipeed Maixuino module, integrated for smart energy-efficiency buildings. Developing ISO standards for PC power monitoring, building on methodologies like ISO/IEC 21836:2020 and IEC 62301, could further enhance its applicability. This research contributes to advancing energy-efficient computing and promoting more sustainable technology practices.

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